

Wireshark Network Traffic Analysis

ICMP & DNS Packet Analysis

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Degree: B.E. – IoT with Cybersecurity and Blockchain Technology

Operating System: Kali Linux

Tool: Wireshark

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1. ICMP Traffic Analysis

The objective was to capture and analyze ICMP traffic using Wireshark. ICMP traffic was generated using the ping command.

Wireshark Filter: `icmp`

- ICMP Echo Request was observed.
- ICMP Echo Reply was observed.
- Source and destination IP addresses were identified.
- IPv4 header information was inspected.
- TTL and packet length were analyzed.



Figure 1: ICMP packet analysis in Wireshark



Figure 2: ICMP response packet

2. DNS Traffic Analysis

DNS traffic was captured and analyzed using Wireshark to understand domain-name resolution.

Wireshark Filter: `dns`

- DNS Standard Query was observed.
- DNS Standard Query Response was observed.

- A and AAAA records were observed.
- Source and destination addresses were inspected.



Figure 3: DNS traffic filtered in Wireshark



Figure 4: DNS response packet analysis

3. Learning Outcomes

- Captured network traffic using Wireshark.
- Used Wireshark display filters.
- Analyzed ICMP Echo Request and Reply packets.
- Analyzed basic DNS queries and responses.
- Inspected IPv4 and ICMP packet information.
- Learned to analyze network communication at packet level.

4. Conclusion

This project provided hands-on experience in capturing and analyzing network traffic using Wireshark on Kali Linux. The analysis helped understand ICMP and DNS communication at the packet level.

ICMP_Analysis.png ICMP_Response.png DNS_Analysis.png DNS_Response.png